THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A pin stuck to a magnet can also pick up another pin because: 1

(a) air presses them **(b)** pins have glue
(c) the pin acts as a temporary magnet **(d)** the pin is sticky

2. Which of these is the correct way to hold a compass while taking a reading? 1

(a) in a metal box **(b)** close to a speaker
(c) flat on the palm, away from iron objects **(d)** tilted near a steel desk

3. An iron nail dangling from a magnet picks up another nail. The second nail is held because: 1

(a) air presses
(b) gravity
(c) the first nail became temporarily magnetised
(d) glue is present

4. Which method would give the strongest magnet from an iron nail? 1

(a) stroking in one direction repeatedly **(b)** painting
(c) stroking back and forth **(d)** heating

5. Which technique best shows that magnetic force is non-contact? 1

(a) the pin is round
(b) the pin moves toward a magnet held at a distance
(c) the pin sticks after touching
(d) the magnet is heavy

6. After removing the magnet, the chain of nails falls apart. This shows the nails: 1

(a) were heavy **(b)** were plastic
(c) were permanent magnets **(d)** were temporary magnets

7. Which graph-like description fits the pull of a magnet on a pin as distance increases? 1

(a) pull increases **(b)** pull becomes infinite
(c) pull stays same **(d)** pull decreases rapidly

8. A student hangs two bar magnets side by side with N facing N. They move apart. What does this show? 1

(a) gravity **(b)** no force
(c) repulsion between like poles **(d)** attraction

9. **Assertion (A):** Like poles of two magnets attract each other. 1
Reason (R): A north pole and a south pole attract each other.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A wooden ruler is attracted by a magnet. 1
Reason (R): Wood is a non-magnetic material.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do ships carry more than one compass? 2

12. Why does a magnet pick up nails but not a wooden pencil? 2

13. How would you test whether a nail is a permanent magnet or a temporary one? 2

14. Describe how a magnet helps in removing iron splinters from the eye safely. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the behaviour of a magnetic needle when a magnet is placed to the east of it. 3

16. Three coins look the same. One is iron, one copper, one aluminium. How would you identify the iron? 3

17. Why is it unsafe to have powerful magnets near children? 3

OR

Explain how a refrigerator magnet and a magnetic board work. 3

18. Why do some magnets have a coating? 3

OR

Compare the magnetic force and the force of gravity on a pin near a magnet. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a lab, Ishita brought a magnet near objects: an iron key, a copper coin, a plastic scale, a steel pin, a glass marble and a nickel piece. Only some of them stuck to the magnet. 5
(i) Name two objects that were attracted.
(ii) Name two objects that were not attracted.
(iii) What are materials attracted by a magnet called?
(iv) Ishita wants to separate iron nails from a heap of sand. Suggest a method and explain.

20. Discuss the importance of magnets in modern life and give examples from home, hospital and industry. 5

OR

Discuss how we know the Earth has a magnetic field using three pieces of evidence. 5

END OF QUESTION PAPER — Total: 40 Marks